

Fig. 3: Prediction of top three demand-generating categories

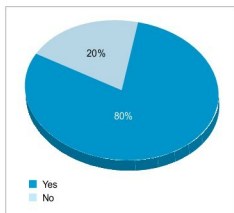


Fig. 4: Demand-supply gap in the market

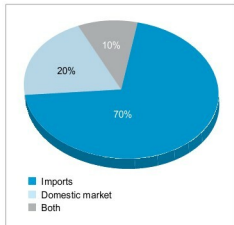


Fig. 5: Major sourcing channels for electronics chemicals and materials

icals and materials, and are widely used as the base for the manufacture of semiconductors. Whereas, PCB laminates are widely used for manufacturing PCBs.

Low-K dielectrics are a relatively recent application of semiconductors and integrated circuits, and therefore are expected to grow aggressively in the next five years. However, the Indian market may be slightly different from the global market with respect to demand.

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The Indian market for electronics chemicals and materials does not, typically, follow global trends. Here, it is primarily driven by silicon wafers, PCB laminates, specialty gases and wet chemicals. Environmentally-safe products will continue to be in demand. Survey participants unanimously predict a huge demand for wet chemicals. According to them, future growth will mainly come from the following three demand-generating categories (Fig. 3):

1. Wet chemicals
2. PCB laminates
3. Silicon wafers

Moreover, 30 per cent survey participants also indicate demand for specialty gases.

Supply side

The Indian market for electronics chemicals and materials has been growing at a fast pace, primarily driven by imports, as local production is insignificant in the country. Eighty per cent of survey participants report a demand-supply gap with respect to local production (Fig. 4). Value addition through local manufacturers in India is quite limited, as a majority of high-value and critical category chemicals are still imported.

Seventy per cent participants mentioned that imports are the main source of electronics chemicals and materials. However, 20 per cent depend on local production,

and the remaining 10 per cent explore both sources (Fig. 5).

The local production ecosystem of electronics chemicals and materials is still not mature. Therefore currently, local sourcing is primarily limited to:

1. Adhesives
2. Basic-level PCB cleaning and coating solutions
3. Solder pastes

Silicon resins are imported in bulk. Advanced conformal coatings are also sourced mainly through imports. According to survey participants, a huge demand-supply gap exists in this category, and cost-effectiveness plays a big role in the market.

Chinese and Korean products have captured a major share of the market. PCB chemicals and fluxes are also imported, as no manufacturing is done in the country.

Moving forward

The Indian electronics chemicals and materials market is highly fragmented owing to the presence of many vendors. The market is characterised by the presence of several well-established global, regional and local suppliers. Competition in the market is expected to intensify due to the growth of the electronics manufacturing industry and technological advances.

Considering the growing importance of this market, a few foreign companies also plan to set up production units in India. This could help bridge the demand-supply gap in this sector. Given that there is immense domestic demand for electronic products, development of the electronics chemicals and materials industry will also help the entire electronics manufacturing ecosystem to achieve long-term sustainable growth in line with 'Make in India' initiative. **EFY**